

Self-RAG: Learning to Retrieve, Generate and Critique through Self-Reflection

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Abstract

Despite their remarkable capabilities, large language models often produce responses that are not supported by facts, especially on knowledge-intensive tasks. We introduce Self-RAG, a framework that trains a language model to adaptively retrieve passages on demand and to reflect on its own generations using special reflection tokens.

1. Introduction

Retrieval-augmented generation methods such as RAG and Fusion-in-Decoder have been shown to improve factual accuracy by conditioning generation on retrieved passages. We build upon this line of work.

However, these methods retrieve a fixed number of passages regardless of whether retrieval is actually needed. This wastes computation on easy queries and can also hurt quality when the retrieved passages are irrelevant.

Unlike prior retrieval-augmented approaches that always retrieve a fixed number of passages, Self-RAG learns to decide when to retrieve.

2. Method

Self-RAG trains a single language model to generate special reflection tokens that indicate whether retrieval is needed, whether a retrieved passage is relevant, and whether the generated text is supported by the evidence.

Instead of always using a fixed retrieval step, the model adaptively decides to retrieve on demand, replacing the fixed retrieval stage used in RAG with a learned decision.

We combine ideas from retrieval-augmented generation and critique-based decoding, integrating them into a single end-to-end trained model.

3. Experiments

Self-RAG outperforms ChatGPT and retrieval-augmented baselines such as RAG on several knowledge-intensive benchmarks.

Ablation studies show that removing the retrieval decision mechanism and always retrieving reduces efficiency without improving accuracy.

We find that the reflection tokens are important: removing them degrades both factual precision and citation accuracy.

4. Limitations

Self-RAG requires training with reflection tokens, which needs additional annotated data and substantial compute.

Adaptive retrieval introduces variable inference cost, which complicates serving in production systems.